

## 4 Problems

**1.** Besides the approach in Section 1.2, there are several alternate ways of dealing with the sum  $\sum 1/(z + \omega)^2$ , where  $\omega = n + m\tau$ . For example, one may sum either (a) circularly, (b) first in  $n$  then in  $m$ , (c) or first in  $m$  then in  $n$ .

(a) Prove that if  $z \notin \Lambda$ , then

$$\lim_{R \rightarrow \infty} \sum_{n^2 + m^2 \leq R^2} \frac{1}{(z + n + m\tau)^2} = S_1(z)$$

exists and  $S_1(z) = \wp(z) + c_1$ .

(b) Similarly,

$$\sum_m \left( \sum_n \frac{1}{(z + n + m\tau)^2} \right) = S_2(z)$$

exists and  $S_2(z) = \wp(z) + c_2$ , where  $c_2 = F(\tau)$ , and  $F$  is the forbidden Eisenstein series.

(c) Also

$$\sum_n \left( \sum_m \frac{1}{(z + n + m\tau)^2} \right) = S_3(z)$$

exists with  $S_3(z) = \wp(z) + c_3$ , and  $c_3 = \tilde{F}(\tau)$ , the reverse of  $F$ .

[Hint: To prove (a), it suffices to show that  $\lim_{R \rightarrow \infty} \sum_{1 \leq n^2 + m^2 \leq R^2} 1/(n + m\tau)^2 = c_1$  exists. This is proved by a comparison with  $\int_{1 \leq x^2 + y^2 \leq R^2} \frac{dx}{(x + y\tau)^2} = I(R)$ . It can be shown that  $I(R) = 0$ , which follows because  $(x + y\tau)^{-2} = -(\partial/\partial x)(x + y\tau)^{-1}$ .]

**2.** Show that

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z + m\tau)\pi)}$$

where  $c$  is an appropriate constant. In fact, by part (b) of the previous problem  $c = -F(\tau)$ .

**3.\*** Suppose  $\Omega$  is a simply connected domain that excludes the three roots of the polynomial  $4z^3 - g_2z - g_3$ . For  $\omega_0 \in \Omega$  and  $\omega_0$  fixed, define the function  $I$  on  $\Omega$  by

$$I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2z - g_3}} \quad \omega \in \Omega.$$

Then the function  $I$  has an inverse given by  $\wp(z + \alpha)$  for some constant  $\alpha$ ; that is,

$$I(\wp(z + \alpha)) = z$$

for appropriate  $\alpha$ .

[Hint: Prove that  $(I(\wp(z + \alpha)))' = \pm 1$ , and use the fact that  $\wp$  is even.]

4.\* Suppose  $\tau$  is purely imaginary, say  $\tau = it$  with  $t > 0$ . Consider the division of the complex plane into congruent rectangles obtained by considering the lines  $x = n/2$ ,  $y = tm/2$  as  $n$  and  $m$  range over the integers. (An example is the rectangle whose vertices are  $0, 1/2, 1/2 + \tau/2$ , and  $\tau/2$ .)

- (a) Show that  $\wp$  is real-valued on all these lines, and hence on the boundaries of all these rectangles.
- (b) Prove that  $\wp$  maps the interior of each rectangle conformally to the upper (or lower) half-plane.

# 10 Applications of Theta Functions

The problem of the representation of an integer  $n$  as the sum of a given number  $k$  of integral squares is one of the most celebrated in the theory of numbers. Its history may be traced back to Diophantus, but begins effectively with Girard's (or Fermat's) theorem that a prime  $4m + 1$  is the sum of two squares. Almost every arithmetician of note since Fermat has contributed to the solution of the problem, and it has its puzzles for us still.

*G. H. Hardy, 1940*

This chapter is devoted to a closer look at the theory of theta functions and some of its applications to combinatorics and number theory.

The theta function is given by the series

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z},$$

which converges for all  $z \in \mathbb{C}$ , and  $\tau$  in the upper half-plane.

A remarkable feature of the theta function is its dual nature. When viewed as a function of  $z$ , we see it in the arena of elliptic functions, since  $\Theta$  is periodic with period 1 and “quasi-period”  $\tau$ . When considered as a function of  $\tau$ ,  $\Theta$  reveals its modular nature and close connection with the partition function and the problem of representation of integers as sums of squares.

The two main tools allowing us to exploit these links are the triple-product for  $\Theta$  and its transformation law. Once we have proved these theorems, we give a brief introduction to the connection with partitions, and then pass to proofs of the celebrated theorems about representation of integers as sums of two or four squares.